



Graph-GRPO: Training Graph Flow Models with Reinforcement Learning

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Motivation

Goal: steer graph flow models (GFM) toward verifiable rewards (validity, docking score, drug-likeness) with online RL.

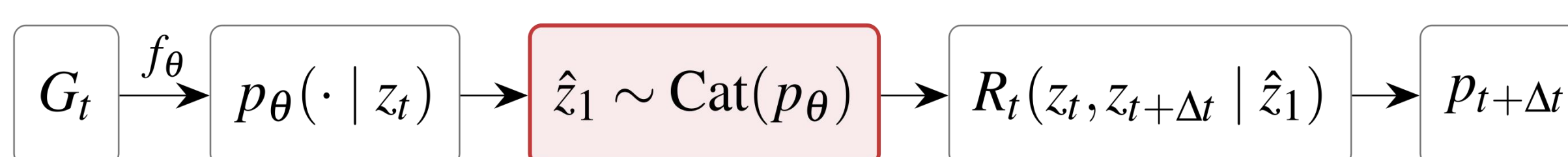
Why it's hard:

Broken gradients. GFM transitions come from non-differentiable Monte-Carlo sampling, but policy gradients need differentiable ones.

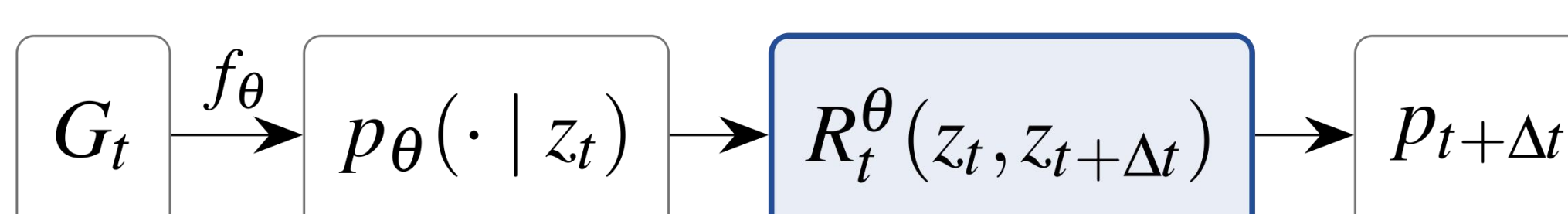
Sparse rewards. Most de-novo graphs are invalid, giving almost no useful signal.

Analytic Transition

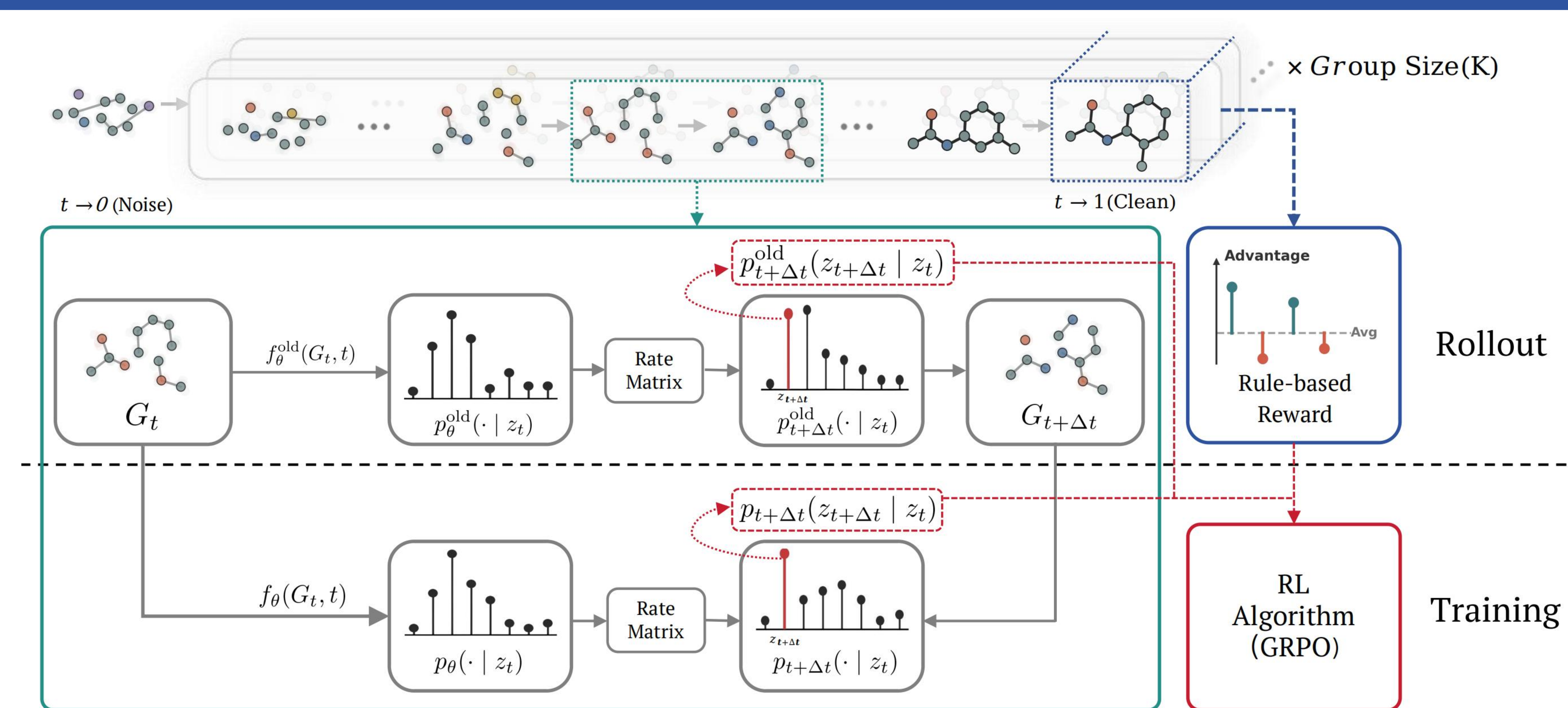
DeFoG: Monte-Carlo sampling, non-differentiable ✗



Graph-GRPO: closed-form, differentiable ✓

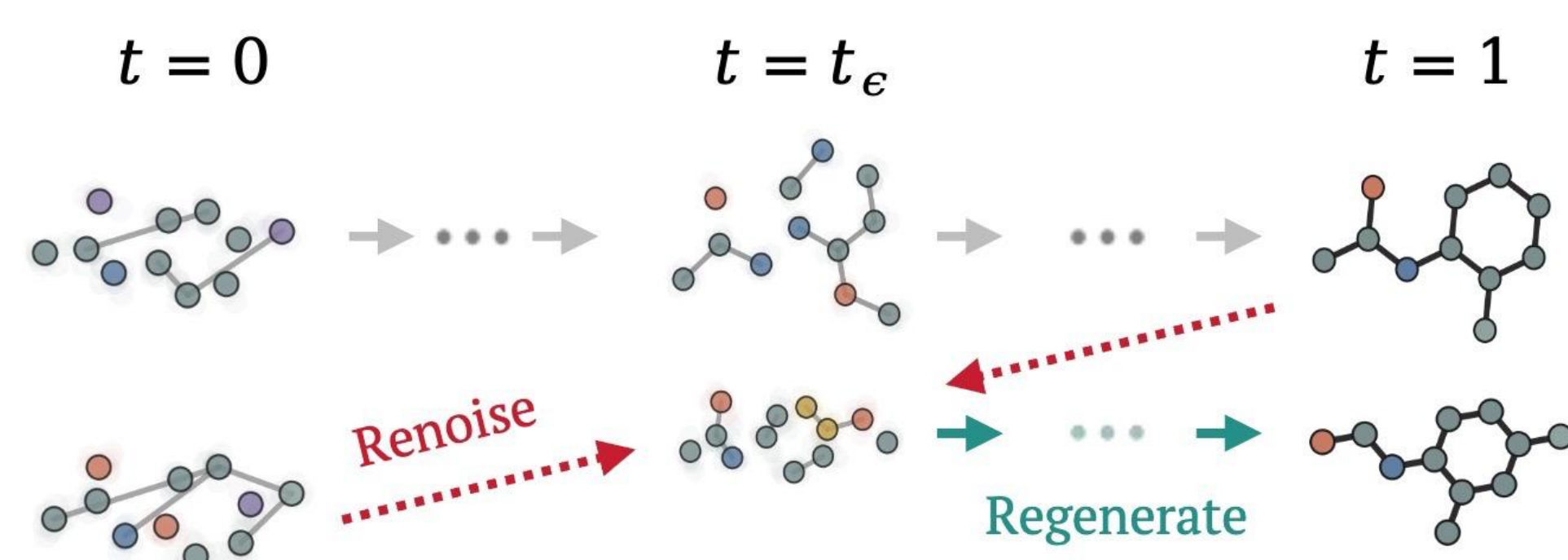


Method: Graph-GRPO



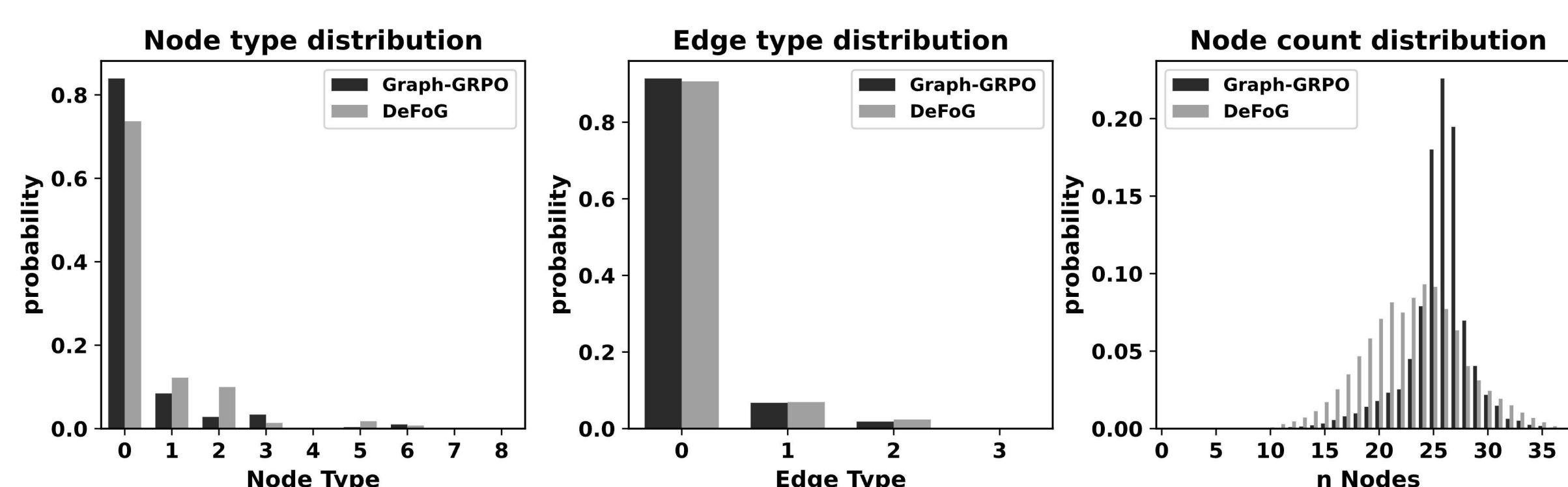
Refinement

Renoise a top candidate back to a small noise level ϵ , then regenerate and keep the better result, locally refining good graphs instead of starting from scratch.



Dynamic Prior

Graph-GRPO adapts its noise prior so the node-type, edge-type, and graph-size distributions concentrate on high-reward regions, unlike fixed prior.



Experiment

PMO Benchmark

Optimizing 23 molecular-property tasks, scored by AUC of the top-10 molecules (with and without a prescreening corpus)

Oracle	w/ Prescreening				Cold-Start (w/o Prescreening)				
	Graph-GRPO	InVirtuoGen	GenMol	f-RAG	Graph-GRPO	InVirtuoGen	Gen. GFN	Mol GA	REINVENT
Albuterol Similarity	0.994	0.975	0.937	0.977	0.994	0.950	0.949	0.896	0.882
Amlodipine MPO	0.823	0.836	0.810	0.749	0.823	0.733	0.761	0.688	0.635
Celecoxib Rediscovery	0.890	0.839	0.826	0.778	0.890	0.798	0.802	0.567	0.713
Deco Hop	0.942	0.968	0.960	0.936	0.762	0.748	0.733	0.649	0.666
DRD2	0.995	0.995	0.995	0.992	0.995	0.985	0.974	0.936	0.945
Fexofenadine MPO	0.984	0.904	0.894	0.856	0.984	0.845	0.856	0.825	0.784
GSK3b	0.948	0.988	0.986	0.969	0.965	0.952	0.881	0.843	0.865
Isomers C7H8N2O2	0.995	0.988	0.942	0.955	0.995	0.968	0.969	0.878	0.852
Isomers C9H10N2O2PF2Cl	0.932	0.898	0.833	0.850	0.932	0.874	0.897	0.865	0.642
JNK3	0.910	0.898	0.906	0.904	0.910	0.825	0.764	0.702	0.783
Median 1	0.388	0.386	0.398	0.340	0.388	0.342	0.379	0.257	0.356
Median 2	0.322	0.377	0.359	0.323	0.300	0.288	0.294	0.301	0.276
Mestranol Similarity	0.980	0.991	0.982	0.671	0.974	0.797	0.708	0.591	0.618
Osimertinib MPO	0.924	0.881	0.876	0.866	0.922	0.870	0.860	0.844	0.837
Perindopril MPO	0.690	0.753	0.718	0.681	0.689	0.645	0.595	0.547	0.537
QED	0.944	0.943	0.942	0.939	0.944	0.942	0.942	0.941	0.941
Ranolazine MPO	0.928	0.854	0.821	0.820	0.928	0.848	0.819	0.804	0.760
Scaffold Hop	0.711	0.711	0.628	0.576	0.622	0.589	0.615	0.527	0.560
Sitagliptin MPO	0.879	0.743	0.584	0.601	0.879	0.709	0.634	0.582	0.021
Thiothixene Rediscovery	0.842	0.652	0.692	0.584	0.842	0.625	0.583	0.519	0.534
Troglitazone Rediscovery	0.711	0.853	0.867	0.448	0.711	0.595	0.511	0.427	0.441
Valsartan SMARTS	0.841	0.935	0.822	0.627	0.841	0.210	0.135	0.000	0.178
Zaleplon MPO	0.697	0.624	0.584	0.486	0.697	0.536	0.552	0.519	0.358
Sum (AUC-top10)	19.270	18.993	18.362	16.928	18.987	16.676	16.213	14.708	14.184

Protein Docking

Designing molecules that bind five target proteins, measured by docking score and hit ratio.

Target	Metric	RL-optimized Generative Models						Basic Generative Models		
		Graph-GRPO	GDPO	DDPO	FREED	REINVENT	GCPN	DiGress-G	DiGress	MOOD
parp1	DS (top 5%) ↓	-12.515	-10.938	-9.247	-10.579	-8.702	-8.102	-9.463	-9.219	-10.865
	Hit Ratio ↑	60.763	9.814	0.419	4.627	0.480	0	1.172	0.366	7.017
fa7	DS (top 5%) ↓	-9.099	-8.691	-7.739	-8.378	-7.205	-6.688	-7.318	-7.736	-8.160
	Hit Ratio ↑	9.367	3.449	0.342	1.332	0.213	0	0.321	0.182	0.733
5ht1b	DS (top 5%) ↓	-11.399	-11.304	-9.488	-10.714	-8.770	-8.544	-8.971	-9.280	-11.145
	Hit Ratio ↑	46.634	34.359	5.488	16.767	2.453	1.455	2.821	4.236	18.673
braf	DS (top 5%) ↓	-11.141	-11.197	-9.470	-10.561	-8.392	-8.713	-8.825	-9.052	-11.063
	Hit Ratio ↑	10.041	9.039	0.445	2.940	0.127	0	0.152	0.122	5.240
jak2	DS (top 5%) ↓	-11.123	-10.183	-8.990	-9.735	-8.165	-8.073	-8.360	-8.706	-10.147
	Hit Ratio ↑	52.897	13.405	1.717	5.800	0.613	0	0.311	0.861	9.200

Ablation Study

How much each component (RL, refinement, pre-screening, dynamic prior) contributes to the final AUC.

Method	RL	Refine-ment	Pre-screening	Dynamic Prior	AUC-top10
DeFoG	-	-	-	-	11.079
DeFoG	-	✓	-	-	15.251
DeFoG	-	✓	✓	-	15.887
Graph-GRPO	✓	-	-	-	16.901
Graph-GRPO	✓	-	-	✓	17.450
Graph-GRPO	✓	✓	-	-	17.950
Graph-GRPO	✓	✓	-	✓	18.987
Graph-GRPO	✓	✓	✓	-	18.033
Graph-GRPO	✓	✓	✓	✓	19.270

Synthetic Graphs

Generating valid, novel Planar and Tree graphs, measured by V.U.N.

Model	Step	Planar		Tree	
		V.U.N. ↑	Ratio ↓	V.U.N. ↑	Ratio ↓
Train set	-	100	1.0	100	1.0
GraphRNN	-	0.0	490.2	0.0	607.0
BiGG	-	5.0	16.0	75.0	5.2
GraphGen	-	7.5	210.3	95.0	33.2
CatFlow	-	80.0	-	-	-
DiGress	1,000	77.5	5.1	90.0	1.6
GBD	1,000	92.5	7.7	-	-
DisCo	1,000	83.6±2.1	-	-	-
GDPO	1,000	73.8±2.5	-	-	-
DeFoG	50	95.0±3.2	3.2±1.1	73.5±9.0	2.5±1.0
Graph-GRPO	50	95.0±4.0	1.5±0.5	97.5±1.6	2.2±0.8